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Reg. No.: 192324189

Course Code: DSA0405

Slot: D

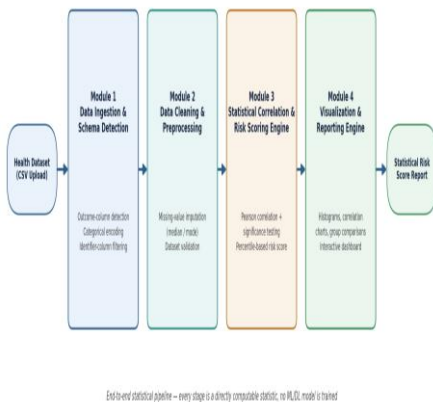
Course Name: Fundamentals of data science

Course Faculty: Dr. Krishna raj Mony

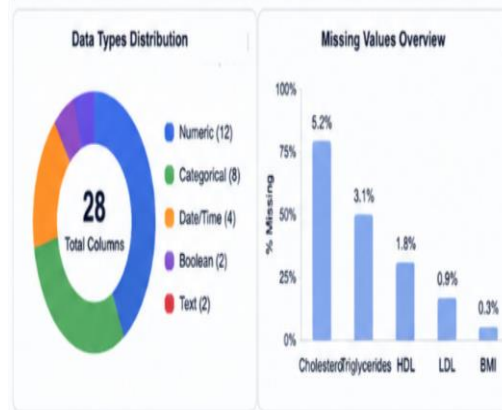
Project Title: Exploratory Statistical Analysis of Clinical and Lifestyle Indicators for Early Health Risk

Module Photographs:

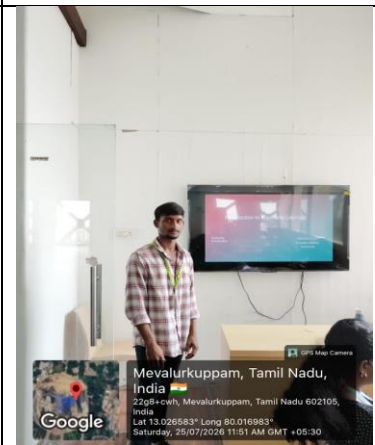
Architecture Diagram



Module1: Data Ingestion & Schema Detection



Presentation Image



Project Description:

Data Ingestion & Schema Detection: This module focuses on collecting, importing, and preparing clinical and lifestyle datasets for further statistical analysis. The module accepts health-related data from sources such as CSV files, Excel spreadsheets, databases, and other structured data sources. It performs data validation, preprocessing, and profiling to ensure that the dataset is suitable for analysis.

The module automatically detects the schema of the dataset by identifying column names, data types, categorical and numerical variables, date/time fields, missing values, duplicate records, and possible relationships between variables. Important clinical and lifestyle indicators such as age, gender, BMI, blood pressure, cholesterol level, glucose level, smoking habits, alcohol consumption, physical activity, sleep duration, and dietary habits are identified and organized. The detected schema and data-quality information provide a structured foundation for subsequent exploratory statistical analysis, correlation analysis, hypothesis testing, regression analysis, and visualization.

Student Signature

Guide Signature